Contents

Part I
preliminaries

1. Basics of Magnetism

1.1 Direct Exchange

1.1.1 Coulomb exchange

We are looking for a mechanism that can explain magnetic ordering at high temperatures. One might expect that this requires a spin-dependent interaction, however, no such interaction of this strength is known. This is where direct exchange enters the stage. It is made up of multiple mechanics, the first one (canonically) being Coulomb exchange. This idea goes back to Heisenberg. It is able to explain how ferromagnetic order comes about in a system of localized spins (e.g. spins of unpaired valence electrons). More precise? The most remarkable thing: the story starts with the (spin-indepedent!) Coulomb interaction between the electrons. In second quantization the Coulomb interaction reads

$$H_{\mathrm{C}} = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\int d^3r_1 \int d^3r_2 \sum_{\sigma_1,\sigma_2} \Psi^\dagger_{\sigma_1}(\vec{r}_1)\Psi^\dagger_{\sigma_2}(\vec{r}_2)\frac{1}{|\vec{r}_1-\vec{r}_2|}\Psi_{\sigma_2}(\vec{r}_2)\Psi_{\sigma_1}(\vec{r}_1). \tag{1}$$

$\Psi^\dagger_{\sigma_1}(\vec{r}_1)$ is a field operator that creates an electron with spin σ_1 at position $\vec{r}_1$. These are now expanded in orthonormal wavefunctions localized around the ions (Wannier functions) $\phi_{\vec{R},m}(\vec{r})$ and spinors χ_σ:

$$\Psi^\dagger_{\sigma_1}(\vec{r}_1) = \sum_{\vec{R},m} a^\dagger_{\vec{R}m\sigma_1}\phi_{\vec{R}m}(\vec{r}_1)\chi_{\sigma_1}. \tag{2}$$

The sum over $\vec{R}$ goes over the positions of all Ions. m counts all angular degrees of freedom, but not the spin. $a^\dagger_{\vec{R},m,\sigma_1}$ is a fermionic creation operator. One finds

$$\begin{aligned} H_{\mathrm{C}} = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0} & \sum_{\substack{\vec{R}^{(1)}\ldots\vec{R}^{(4)}\\ m^{(1)}\ldots m^{(4)}}} \int d^3r_1 \int d^3r_2\, \phi^*_{\vec{R}^{(1)}m^{(1)}}(\vec{r}_1)\phi^*_{\vec{R}^{(2)}m^{(2)}}(\vec{r}_2)\frac{1}{|\vec{r}_1-\vec{r}_2|}\phi_{\vec{R}^{(3)}m^{(3)}}(\vec{r}_2)\phi_{\vec{R}^{(4)}m^{(4)}}(\vec{r}_1) \\ & \times \sum_{\sigma_1,\sigma_2} \chi^\dagger_{\sigma_1}\chi^\dagger_{\sigma_2}\chi_{\sigma_2}\chi_{\sigma_1} a^\dagger_{\vec{R}^{(1)}m^{(1)}\sigma_1} a^\dagger_{\vec{R}^{(2)}m^{(2)}\sigma_2} a_{\vec{R}^{(3)}m^{(3)}\sigma_2} a_{\vec{R}^{(4)}m^{(4)}\sigma_1}. \end{aligned} \tag{3}$$

The expression is simplified by evaluating the spinor product.

$$\begin{aligned} H_{\mathrm{C}} = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0} \sum_{\substack{\vec{R}^{(1)}\ldots\vec{R}^{(4)}\\ m^{(1)}\ldots m^{(4)}\\ \sigma_1,\sigma_2}} & \left\langle \vec{R}^{(1)}m^{(1)},\vec{R}^{(2)}m^{(2)} \middle| \frac{1}{|\vec{r}_1-\vec{r}_2|} \middle| \vec{R}^{(3)}m^{(3)},\vec{R}^{(4)}m^{(4)} \right\rangle \\ & \times a^\dagger_{\vec{R}^{(1)}m^{(1)}\sigma_1} a^\dagger_{\vec{R}^{(2)}m^{(2)}\sigma_2} a_{\vec{R}^{(3)}m^{(3)}\sigma_2} a_{\vec{R}^{(4)}m^{(4)}\sigma_1} \end{aligned} \tag{4}$$

It is worthwile to take a few detours in order to explore different effects of the Coulomb interaction, increasing the complexity in the mean-time.

on-site, single orbital: Hubbard U-term

First, focus on the on-site effects $\vec{R}^{(1)} = \vec{R}^{(2)} = \vec{R}^{(3)} = \vec{R}^{(4)}$, dropping the position quantum numbers. The easiest case is that of a single orbital per site, $m^{(1)} = m^{(2)} = m^{(3)} = m^{(4)} \equiv m$.

$$\Rightarrow H_{\mathrm{C}} = \frac{1}{2}\sum_{\vec{R}}\sum_{\sigma_1\sigma_2}\left\langle \frac{1}{|\vec{r}_1-\vec{r}_2|}\right\rangle_{m,m} a^\dagger_{\sigma_1}a^\dagger_{\sigma_2}a_{\sigma_2}a_{\sigma_1} \tag{5}$$

The position indices of the ladder operators have been supressed.

$$\sum_{\sigma_1\sigma_2} a^\dagger_{\sigma_1} a^\dagger_{\sigma_2} a_{\sigma_2} a_{\sigma_1} = a^\dagger_\downarrow a^\dagger_\uparrow a_\uparrow a_\downarrow + a^\dagger_\uparrow a^\dagger_\downarrow a_\downarrow a_\uparrow = 2 a^\dagger_\downarrow a^\dagger_\uparrow a_\uparrow a_\downarrow \tag{6}$$

Two out of the four terms ($\sigma_1 = \sigma_2 =\uparrow$ and $\sigma_1 = \sigma_2 =\downarrow$) are identically zero, since $(a_{\downarrow/\uparrow})^2 = 0$. In other words: due to Pauli principle the spin allignement is already fixed. So no exchange interaction can arise here. The technical reason is that we assumed a single orbital on a single site (2 states), leaving the electrons no other choice. We'll have to loosen the approximations to see exchange play out. Let's close with one last observation regarding the one-site, single orbital story;

$$\Rightarrow H_{\mathrm{C}} = \sum_{\vec{R}} U a^\dagger_\downarrow a^\dagger_\uparrow a_\uparrow a_\downarrow, \quad U = \frac{e^2}{4\pi\varepsilon_0} \int d^3 r_1 \int d^3 r_2 \, |\phi(\vec{r}_1)|^2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\phi(\vec{r}_2)|^2 \tag{7}$$

This gives the Hubbard U-term. It's interpretation is pretty straight forward: electrons occupying the same orbital have increased Coulomb energy, given by U. This presents a convenient way to include Coulomb interaction in tight-binding models. It is only an approximation, since all other terms (multiple sites, different orbitals) have been ignored so far. It is also clear that this contribution is the largest off all effects arising from the Coulomb interaction as the overlap is biggest here.

on-site, multiple orbitals: Hund's rule

The first step of complexification is to allow multiple orbitals at every ion, thereby reintroducing the $m^{(i)}$ quantum numbers. If one treats the Coulomb interaction perturbatively, in first order the creation/ annihilation operators have to be grouped into pairs in order to produce non-vanishing contributions. I am not satisfied with this. There are two possibilities: $m^{(1)} = m^{(3)}$, $m^{(2)} = m^{(4)}$ and $m^{(1)} = m^{(4)}$, $m^{(2)} = m^{(3)}$.

$$\begin{aligned} K_{m^{(1)}m^{(2)}} &:= \frac{e^2}{4\pi\varepsilon_0} \left\langle m^{(1)}, m^{(2)} \middle| \frac{1}{|\vec{r}_1 - \vec{r}_2|} \middle| m^{(2)}, m^{(1)} \right\rangle \\ &= \frac{e^2}{4\pi\varepsilon_0} \int d^3 r_1 \int d^3 r_2 \, |\phi_{m^{(1)}}(\vec{r}_1)|^2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\phi_{m^{(2)}}(\vec{r}_2)|^2 \end{aligned} \tag{8}$$

$$\begin{aligned} J_{m^{(1)}m^{(2)}} &:= \frac{e^2}{4\pi\varepsilon_0} \left\langle m^{(1)}, m^{(2)} \middle| \frac{1}{|\vec{r}_1 - \vec{r}_2|} \middle| m^{(1)}, m^{(2)} \right\rangle \\ &= \frac{e^2}{4\pi\varepsilon_0} \int d^3 r_1 \int d^3 r_2 \, \phi^*_{m^{(1)}}(\vec{r}_1) \phi^*_{m^{(2)}}(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \phi_{m^{(1)}}(\vec{r}_2) \phi_{m^{(2)}}(\vec{r}_1) \end{aligned} \tag{9}$$

$$\begin{aligned} \Rightarrow H_C \approx \frac{1}{2} \sum_{\vec{R}} \sum_{\substack{m^{(1)}, m^{(2)} \\ \sigma_1, \sigma_2}} \Big(& K_{m^{(1)}m^{(2)}} a^\dagger_{m^{(1)}\sigma_1} a^\dagger_{m^{(2)}\sigma_2} a_{m^{(2)}\sigma_2} a_{m^{(1)}\sigma_1} \\ & + J_{m^{(1)}m^{(2)}} a^\dagger_{m^{(1)}\sigma_1} a^\dagger_{m^{(2)}\sigma_2} a_{m^{(1)}\sigma_2} a_{m^{(2)}\sigma_1} \Big) \end{aligned} \tag{10}$$

The position indices have been supressed! The exchange term may be reexpressed as

$$\begin{aligned} -\sum_{\vec{R}} \sum_{m^{(1)}, m^{(2)}} J_{m^{(1)}m^{(2)}} \Big(& \left(a^\dagger_{m^{(1)}\uparrow} a_{m^{(1)}\uparrow} - a^\dagger_{m^{(1)}\downarrow} a_{m^{(1)}\downarrow} \right) \left(a^\dagger_{m^{(2)}\uparrow} a_{m^{(2)}\uparrow} - a^\dagger_{m^{(2)}\downarrow} a_{m^{(2)}\downarrow} \right) \\ & + a^\dagger_{m^{(1)}\uparrow} a_{m^{(1)}\downarrow} a^\dagger_{m^{(2)}\downarrow} a_{m^{(2)}\uparrow} + a^\dagger_{m^{(1)}\downarrow} a_{m^{(1)}\uparrow} a^\dagger_{m^{(2)}\uparrow} a_{m^{(2)}\downarrow} \\ & + a^\dagger_{m^{(1)}\uparrow} a_{m^{(1)}\uparrow} a^\dagger_{m^{(2)}\downarrow} a_{m^{(2)}\downarrow} + a^\dagger_{m^{(1)}\downarrow} a_{m^{(1)}\downarrow} a^\dagger_{m^{(2)}\uparrow} a_{m^{(2)}\uparrow} \Big). \end{aligned} \tag{11}$$

Technically there are terms missing from the Fermi-algebra. However, these are only bilinear and thus act as a siengle-particle potential and are not relevant here. With the mapping/ substitution

$$n_m = a^\dagger_{m\uparrow}a_{m\uparrow} + a^\dagger_{m\downarrow}a_{m\downarrow}, \quad s_{mz} = \frac{1}{2}\left(a^\dagger_{m\uparrow}a_{m\uparrow} - a^\dagger_{m\downarrow}a_{m\downarrow}\right), \quad s_{m+} = a^\dagger_{m\uparrow}a_{m\downarrow} \tag{12}$$

one ends up with

$$H_\mathrm{C} = \frac{1}{2}\sum_{\vec{R}}\sum_{m^{(1)},m^{(2)}}\left(K_{m^{(1)}m^{(2)}} - \frac{1}{2}J_{m^{(1)}m^{(2)}}\right)n_{m^{(1)}}n_{m^{(2)}} - 2J_{m^{(1)}m^{(2)}}\vec{s}_{m^{(1)}}\cdot\vec{s}_{m^{(2)}}. \tag{13}$$

The first term is the on-site Coulomb interaction between electrons in different orbitals on the same site. Additionally, $K_{m^{(1)}m^{(2)}} - \frac{1}{2}J_{m^{(1)}m^{(2)}} > 0$, meaning the Coulomb interaction remains repulsive. Returnung to the exchange term; using

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} = \int \frac{d^3k}{(2\pi)^3}\frac{4\pi}{k^2}e^{i\vec{k}(\vec{r}_1-\vec{r}_2)} \tag{14}$$

$$\Rightarrow J_{m^{(1)}m^{(2)}} = \frac{e^2}{\varepsilon_0}\int\frac{d^3k}{(2\pi)^3}\frac{1}{k^2}\underbrace{\int d^3r_1\ \phi^*_{m^{(1)}}(\vec{r}_1)\phi_{m^{(2)}}(\vec{r}_1)e^{i\vec{k}\vec{r}_1}}_{I(\vec{r})}\underbrace{\int d^3r_2\ \phi_{m^{(1)}}(\vec{r}_2)\phi^*_{m^{(2)}}(\vec{r}_2)e^{-i\vec{k}\vec{r}_2}}_{I^*(\vec{r})}. \tag{15}$$

Hence, $J_{m^{(1)}m^{(2)}}$ is positive. Therefore the interaction is minimized by ferromagnetic allignement. This is the derivation of Hund's first rule: the ground state of a partially filled shell is the one that maximizes $\vec{s}_\mathrm{tot}$.

different site, single orbital: Coulomb exchange

Assuming again singular orbitals $m^{(1)} = m^{(2)} = m^{(3)} = m^{(4)} \equiv m$, whilst reinstating the position quantum numbers the calculation is symmetrical to that in the last subchapter.

$$\Rightarrow H_\mathrm{C} = \frac{1}{2}\sum_{\vec{R}^{(1)},\vec{R}^{(2)}}\left(K_{\vec{R}^{(1)}\vec{R}^{(2)}} - \frac{1}{2}J_{\vec{R}^{(1)}\vec{R}^{(2)}}\right)n_{\vec{R}^{(1)}}n_{\vec{R}^{(2)}} - 2J_{\vec{R}^{(1)}\vec{R}^{(2)}}\vec{s}_{\vec{R}^{(1)}}\cdot\vec{s}_{\vec{R}^{(2)}} \tag{16}$$

$$K_{\vec{R}^{(1)}\vec{R}^{(2)}} = \frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\int d^3r_2\left|\phi_{\vec{R}^{(1)}}(\vec{r}_1)\right|^2\frac{1}{|\vec{r}_1-\vec{r}_2|}\left|\phi_{\vec{R}^{(2)}}(\vec{r}_2)\right|^2 \tag{17}$$

$$J_{\vec{R}^{(1)}\vec{R}^{(2)}} = \frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\int d^3r_2\phi^*_{\vec{R}^{(1)}}(\vec{r}_1)\phi^*_{\vec{R}^{(2)}}(\vec{r}_2)\frac{1}{|\vec{r}_1-\vec{r}_2|}\phi_{\vec{R}^{(1)}}(\vec{r}_2)\phi_{\vec{R}^{(2)}}(\vec{r}_1) \tag{18}$$

As in the last chapter $J_{\vec{R}^{(1)}\vec{R}^{(2)}} > 0$, so again the interaction is ferromagnetic. The interpretation is the same as for the previous subchapter; electrons in a triplet state avoid each other in real space due to Pauli-principle ("exchange hole"), thereby reducing their Coulomb energy.

In terms of explaining magnetic ordering this obviously can't be the end of the story, since we haven't found a way to explain anti-ferromagnetic order yet.

2. Partons

2.1 SU(2) Slave Fermions

The method described here to represent spins in terms of Fermions was first eomployed by Abrikosov in 1965 and later reformulated in terms of gauge theory by Affleck in 1988. The idea (for spin 1/2!) is to introduce two new flavours of fermions: $c^\dagger_{i\uparrow}$, $c^\dagger_{i\downarrow}$ on each site. The transformation reads

$$\vec{S}_i = \frac{1}{2}\sum_{\alpha,\beta} c^\dagger_{i\alpha}\vec{\sigma}_{\alpha\beta}c_{i\beta}, \tag{19}$$

where $\vec{\sigma}$ is the vector of Pauli matrices.

This transformation is quite natural, as it has been found even before Abrikosov, just the other way around. It is known that the large-U limit of the Hubbard model at half-filling gives a antiferromagnetic Heisenberg model. In the perturbation theory one finds the exact representation of Spins in terms of fermionic ladder operators given above.

2.1.1 Hilbert space and un-physical states

The mapping 19 is exact on the operator level, meaning, that the eigenspaces of S_i^2 and S_i^α remain unchanged.

The local Hilbert space, however, gets enlarged. We drop the site-indices from now on. The spin-1/2 Hilbert space is of course a two-dimensional Hilbert space

$$\mathcal{H}_{S=\frac{1}{2}} = \mathrm{span}\left(\left|\uparrow\right\rangle, \left|\downarrow\right\rangle\right) \tag{20}$$

The two-fermion Hilbert space on the other hand is four-dimenstional

$$\mathcal{H}_{\mathrm{fermions}} = \mathrm{span}\left(\left|00\right\rangle, \left|10\right\rangle, \left|01\right\rangle, \left|11\right\rangle\right) \tag{21}$$

The idea of 19 is to identify $\mathcal{H}_{S=\frac{1}{2}}$ with the one-particle sector of $\mathcal{H}_{\mathrm{fermions}}$. This leaves two un-physical states: $\left|00\right\rangle$ and $\left|11\right\rangle$. The spin-operators S^α act trivially on these, which is why everything goes through smoothly on the operator level.

Therefore, as long as the Hamiltonian only consists of spin operators, the un-physical states do not influence any physical results (apart from normalization, maybe). This can not garuanteed anymore when treating the system perturbatively using for example a mean-field theory. For this reason we introduce the constraint

$$\left\langle\psi\right| n_{i\downarrow} + n_{i\uparrow} \left|\psi\right\rangle = 1, \tag{22}$$

which is a constraint on the state $\left|\psi\right\rangle$.

2.1.2 global SU(2) symmetrie

Many important models, e.g. exchange models, have spin rotation symmetrie. The symmetrie group of spin-rotations is SU(2). Rewriting 19 in Matrix notation (for S=1/2, without position indices):

$$\vec{S} = \frac{1}{2}\begin{pmatrix} c^\dagger_\uparrow & c^\dagger_\downarrow \end{pmatrix} \vec{\sigma} \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \tag{23}$$

the spin rotation can be represented by $\vec{\sigma_2} \mapsto V^{-1}\vec{\sigma}V^{-1}$, with V an element of SU(2). The task at hand is to determine, how the spin rotation acts on the ladder operators. One may reinterpret

$$\vec{S} \to \frac{1}{2}\begin{pmatrix} c^\dagger_\uparrow & c^\dagger_\downarrow \end{pmatrix} V^{-1}\vec{\sigma}V \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \tag{24}$$

as

$$\begin{pmatrix} c^\dagger_\uparrow & c^\dagger_\downarrow \end{pmatrix} \to \begin{pmatrix} c^\dagger_\uparrow & c^\dagger_\downarrow \end{pmatrix} V \quad \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \to V^{-1}\begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \quad \vec{\sigma} \to \vec{\sigma} \tag{25}$$

.

$c_\downarrow$, $c_\uparrow$ transform as a SU(2) doublet under left multiplication by an SU(2) matrix. To cosntruct the conjugate doublet, which transforms in the same manner, consider

$$\begin{pmatrix} c_\uparrow^\dagger \\ c_\downarrow^\dagger \end{pmatrix} \to \left[\begin{pmatrix} c_\uparrow^\dagger & c_\downarrow^\dagger \end{pmatrix} V\right]^T = V^T \begin{pmatrix} c_\uparrow^\dagger \\ c_\uparrow^\dagger \end{pmatrix} = V^T \sigma_3 \sigma_3 \begin{pmatrix} c_\uparrow^\dagger \\ c_\uparrow^\dagger \end{pmatrix} \tag{26}$$

To recover V^{-1}, multiply both sides by σ_3 from the left and use $V^{-1} = V^H = \sigma_3 V^T \sigma_3$ (σ_3 represents the complex conjugation in SU(2) better formulation?).

$$\Rightarrow \sigma_3 \begin{pmatrix} c_\uparrow^\dagger \\ c_\downarrow^\dagger \end{pmatrix} \to V^{-1} \sigma_3 \begin{pmatrix} c_\uparrow^\dagger \\ c_\downarrow^\dagger \end{pmatrix} \tag{27}$$

.

Hence, the two conjugate doublets are

$$\begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} c_\uparrow^\dagger \\ -c_\downarrow^\dagger \end{pmatrix} \tag{28}$$

.

It is convention two write these two doublets in matrix:

$$\psi = \begin{pmatrix} c_\uparrow & c_\downarrow \\ c_\downarrow^\dagger & -c_\uparrow \end{pmatrix} \tag{29}$$

2.1.3 local SU(2) symmetrie

The last section described how to recover the two dimenstional spin-1/2 Hilbert-space from the larger four dimensional two-fermion Hilbert-space. It turns out that this is not enough to completely fix the relation between the spin operators S^α and the fermionic operators $c_\updownarrow^{(\dagger)}$. In fact there exist other legitimate mappings besides 19. (19 being arguably the most intuitive of those.)

"legitimate" meaning, that the spin-algebra is left intact:

$$\left[S_i^\alpha, S_j^\beta\right] = i\delta_{ij}\, \varepsilon_{\alpha\beta\gamma}\, S_i^\gamma. \tag{30}$$

3. Mechanics of Continua

3.1 Kinematics

3.1.1 Notation and Coordinates

I will skip over the very technical mathmatical description of the basic objects (like bodies and points) and skip ahead to the interesting stuff.

It is convention to use large latin letters for quantities in the reference configuration and small latin letters for quantities of a particular configuration. One might also refer to the reference configuratoin as the initial configuration.

For example: the position of a particle in the reference configuration is called X, while it is called x in a particular configuration.

This notation gets inherited by all objects in the theory through the variabeles they depend on.

3.1.2 Eulerian and Lagrangian description

When describing how the configuration of a body changes over time it can be advantageous to describe different processes from the points of views of different observers. It comes as no surprise that the choice of description does not change the physics but is merely a matter of convencience. There are two canonical descriptions to choose from;

In the **Eulerian description** (or spatial description) $X(x,t)$ is a function of t and x.

The intutive picture is that of an observer that observes a certain point x in space. Like a hiker resting next to stream, observing it as it flows by. If they encounter particle P at time t (at their point of observation x), they can infer the point $X_P(x,t)$, where P would have been in the reference configuration.

In this sense x is the only free variable besides t. Therefore, quantities in the Eulerian picture are always denoted by small latin letters.

In the **Lagrangian description** (or material description) $x(X,t)$ is a function of t and X.

In this case the observer follows a particle (or a collection of particles, called a region) as it evolves over time. Imagine the hiker built a raft and rides it down the afformentioned stream (assuming the rafts velocity matches that of the stream). Knowing the point X where they started rafting they can reason where they will end up at time t; the point being $x(X,t)$.

Hence, here X and t are the free coordinates. All material quantities are denoted by capital greek letters.

3.1.3 Trajectories and Displacment Field

The object conntecting the afformentioned descriptions is the **trajectorie** $\chi(X,t)$:

$$x = \chi(X,t) \quad \Leftrightarrow \quad X = \chi^{-1}(x,t). \tag{31}$$

It maps out where a observed particle that started at X will be at time t. Obviously, χ itself is part of the Lagrangian description (χ^{-1} belongs to the Eulerian description). A function $f(x,t)$ in Eulerian description can be transformed according to:

$$f(x,t) \mapsto F(X,t) = f(\chi(X,t),t), \tag{32}$$

where F is now in the Lagrangian description. Though f and F are conceptually equivalent they are, mathematically speaking, two different objects.

An example for such a fuunction is the **displacement field**:

$$\begin{aligned} U(X,t) &= X - x(X,t) \\ u(x,t) &= X(x,t) - x \end{aligned} \tag{33}$$

$$U(X,t) \mapsto u(x,t) = U(\chi^{-1}(x,t),t) \tag{34}$$

The displacement field describes how much a particle is displaced compared to the reference configuration.

3.1.4 local changes and Deformation Gradient

Part II
main part

4. Hamiltonian

The Hamiltonian is

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right) \tag{35}$$

Using the Abrikosov representation of the spin in terms of fermions:

$$\vec{S}_i = \sum_{\alpha\beta} f^\dagger_{i\alpha} \frac{\vec{\sigma}}{2} f_{i\beta} \tag{36}$$

$$\begin{aligned} \Rightarrow \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} \sum_{\alpha\beta\gamma\delta} f^\dagger_{i\alpha} f_{i\beta} f^\dagger_{j\gamma} f_{j\delta} \quad \underbrace{\vec{\sigma}_{\alpha\beta} \vec{\sigma}_{\gamma\delta}}_{=\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}} \\ &= \frac{1}{4} \sum_{\alpha\beta} 2 f^\dagger_{i\alpha} f_{i\beta} f^\dagger_{j\beta} f_{j\alpha} - f^\dagger_{i\alpha} f_{i\alpha} f^\dagger_{j\beta} f_{j\beta} \end{aligned}$$

Move the last annihilation operator in the first term to the second postion: use $f^\dagger_{j\beta} f_{j\alpha} = \delta_{\alpha\beta} - f_{j\alpha} f^\dagger_{j\beta}$ and acquire 1 minus signs. After that bring the other annihilation operator to the last position which gives another minus.

$$\vec{S}_i \cdot \vec{S}_j = \frac{1}{4} \sum_{\alpha\beta} -2 f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{j\beta} + 2 f_{i\alpha} f_{i\beta} \delta_{\alpha\beta} - f^\dagger_{i\alpha} f_{i\alpha} f^\dagger_{j\beta} f_{j\beta} \tag{37}$$

The second term is n_i and the third is $n_i n_j$. For the first term I introduce the shorthand

$$\chi_{ij} = \frac{1}{2} \sum_{\alpha} f^\dagger_{i\alpha} f_{j\alpha} \tag{38}$$

$$\vec{S}_i \cdot \vec{S}_j = -2 \chi_{ij} \chi_{ji} + \frac{1}{2} n_i - \frac{1}{4} n_i n_j \tag{39}$$

Using the constaint $n_i = n_j = 1$, the final expression is

$$\vec{S}_i \cdot \vec{S}_j = -2 \chi_{ij} \chi_{ji} + \frac{1}{4} \tag{40}$$

Relevant for the Hamiltonian is also the term:

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 = \frac{1}{16} - \chi_{ij} \chi_{ji} + 4 \left(\chi_{ij} \chi_{ji} \right)^2 \tag{41}$$

The mean field decoupling works like

$$\chi_{ij} \chi_{ji} \approx \langle \chi_{ij} \rangle \chi_{ji} + \langle \chi_{ji} \rangle \chi_{ij} - |\langle \chi_{ij} \rangle|^2 \tag{42}$$

$$\left(\chi_{ij} \chi_{ji} \right)^2 \approx 2 |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + 2 |\langle \chi_{ij} \rangle|^2 \langle \chi_{ji} \rangle \chi_{ij} - 3 |\langle \chi_{ij} \rangle|^4 \tag{43}$$

$$\Rightarrow \vec{S}_i \cdot \vec{S}_j \approx -2\langle\chi_{ij}\rangle\chi_{ji} + \text{h.c.} + 2|\langle\chi_{ij}\rangle|^2 + \frac{1}{4} \tag{44}$$

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j\right)^2 \approx -\langle\chi_{ij}\rangle\chi_{ji} + 8|\langle\chi_{ij}\rangle|^2\langle\chi_{ij}\rangle\chi_{ji} + \text{h.c.} + |\langle\chi_{ij}\rangle|^2 - 12|\langle\chi_{ij}\rangle|^4 + \frac{1}{16} \tag{45}$$

Bringing everything together:

$$\begin{aligned}\mathcal{H} \approx \sum_{\langle ij\rangle} J_{ij}\Bigg[&\left(-1-\kappa/2+4\kappa|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{h.c}\\ &+\left(2+\kappa-12\kappa|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2+\frac{1}{4}+\frac{\kappa}{16}\Bigg]\end{aligned} \tag{46}$$

It is sometimes handy to define

$$\tilde{J}_{ij} = J_{ij}\left(1+\kappa/2\right), \quad \tilde{\kappa} = \kappa/(1+\kappa/2) \tag{47}$$

I drop the constant:

$$\begin{aligned}\mathcal{H} \approx \sum_{\langle ij\rangle} \tilde{J}_{ij}\Bigg[&-\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{h.c}\\ &+2\left(1-6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2\Bigg]\end{aligned} \tag{48}$$

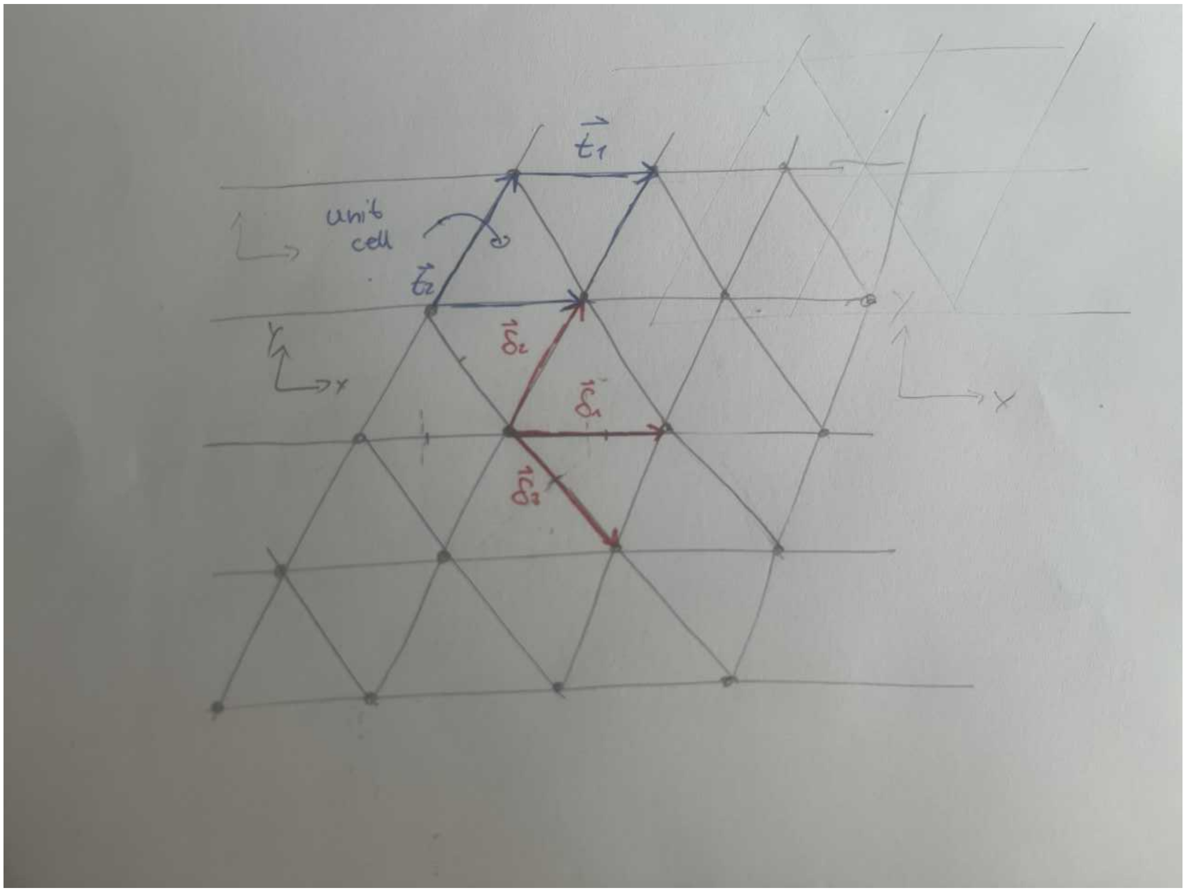
Hence, the mean field produces the effective hopping amplitude t^{MF}:

$$t^{\text{MF}}_{ji} = -\tilde{J}_{ij}\langle\chi_{ji}\rangle\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right) = -J_{ij}\langle\chi_{ij}\rangle\left(1+\kappa/2-4\kappa|\langle\chi_{ij}\rangle|^2\right). \tag{49}$$

$$\Rightarrow \mathcal{H} \approx \sum_{\langle ij\rangle} t^{\text{MF}}_{ji}\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right) + \text{const.} \tag{50}$$

5. Thermodynamic Limit of Unstrained model

5.1 0-0 and π-π States



This system is a simple triangular Bravais lattice with primitve translations

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}. \tag{51}$$

5.1.1 Pedestrian method

The tight-binding Hamiltonian is:

$$\begin{aligned} H &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f^\dagger_{i\sigma} f_{i+\delta\,\sigma} + \text{h.c.} \right) + \mu \sum_i \left(f^\dagger_{i\uparrow} f_{i\uparrow} + f^\dagger_{i\downarrow} f_{i\downarrow} - 1 \right) + \text{const.} \\ &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f^\dagger_{i\sigma} f_{i+\delta\,\sigma} + \text{h.c.} \right) + \mu \sum_{i,\sigma} f^\dagger_{i\sigma} f_{i\sigma} - \mu N + \text{const.}. \end{aligned} \tag{52}$$

i runs over sites sites and δ over nearest neighbours of a generic site. I include the absolute shift for the sake of consistency. The factor 1/2 is also included for consistency. The constant is a product of mean field theory and reads

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta}|^2 \rangle \right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \rangle \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) \tag{53}$$

The second step can be understood in at least two ways; a) each site has 6 neighbours, or b) there are bond per primitive unit cell 1 of which is completely inside the unit cell and contributes a 2 and the other 4 shared between 2 unit cells thereby contributing a 1. All bonds are equivalent of course. The afformentioned six nearest neighbours lie at the (relative) positions

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \ \vec{\delta}_2 = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \ \vec{\delta}_3 = \begin{pmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

and $-\vec{\delta}_1,\ -\vec{\delta}_2,\ -\vec{\delta}_3$. I will start with the $0-0$ state. To advance, do a Fourier transform:

$$f_{i\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{i\vec{k}\vec{R}_i} c_{k\sigma}. \tag{54}$$

$$\begin{aligned}
\Rightarrow H &= \sum_{q,\sigma} \frac{t}{2} \left(c^\dagger_{q\sigma} c_{q\sigma} \left(\sum_\delta e^{i\vec{q}\vec{\delta}} + \mu \right) + \text{h.c.} \right) - \mu N \\
&= \sum_{q,\sigma} \left(t \sum_\delta \cos\left(\vec{q}\vec{\delta}\right) + \mu \right) c^\dagger_{q\sigma} c_{q\sigma} - \mu N
\end{aligned} \tag{55}$$

Using first the symmetrie of the cosine and then the sum identity for the cosine the dispersion reads

$$\begin{aligned}
\varepsilon(\vec{q}) &= 2t \left[\cos(q_x) + \cos\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) + \cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \right] + \mu \\
&= 2t \left[\cos(q_x) + 2\cos\left(\frac{q_x}{2}\right)\cos\left(\frac{\sqrt{3}q_y}{2}\right) \right] + \mu
\end{aligned} \tag{56}$$

features:

- $\varepsilon(\vec{q}) \ \in \ [-3t, 6t] + \mu \Rightarrow \mu \ \in \ [-6t, 3t]$
- symmetries $q_x \to -q_x$ and $q_y \to -q_y$ (+ should have some rotational symmetrie and (thereby associated) inflections along other axes) could check this
- periodicity: 4π in q_x, $(4\pi)/\sqrt{3}$ in q_y

The mean field parameters are

$$\langle \chi_{ij} \rangle = \frac{1}{2} \langle f^\dagger_{i\uparrow} f_{j\uparrow} + f^\dagger_{i\downarrow} f_{j\downarrow} \rangle = \langle f^\dagger_{i\uparrow} f_{j\uparrow} \rangle \tag{57}$$

Simplify the expression by Fourier transform:

$$\begin{aligned}
\langle \chi_{ij} \rangle &= \frac{1}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{k}\vec{R}_i)} \langle c^\dagger_p c_q \rangle \\
&= \frac{1}{N} \sum e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q}))
\end{aligned} \tag{58}$$

Take the thermodynamic limit $\sum_k \to (N\Omega)/(2\pi)^2 \int dk$ (Ω is the unit cell volume).

$$\Rightarrow \langle \chi_{ij} \rangle = \Omega \int\limits_{1.\text{BZ}} \frac{d^2q}{(2\pi)^2} e^{i\vec{q}\left(\vec{R}_j - \vec{R}_i\right)} f(\varepsilon(\vec{q})) \tag{59}$$

It looks like the model allows for different χ_{ij} in each direction $\vec{R}_j - \vec{R}_i$ (there are 3 of those)?

From this expression, recover the on-site particle number by setting $i = j$ and accounting for the factor $1/2$ in the definition of χ_{ij}:

$$\langle N_i \rangle = 2\Omega \int \frac{d^2q}{(2\pi)^2} f(\varepsilon(\vec{q})). \tag{60}$$

By construction the particle number is uniform.
Some elaborations on the integral domain follow. The primite translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \tag{61}$$

I do a basis transformation onto thos axes defined by

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \tag{62}$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \tag{63}$$

$$\Rightarrow \det J = -2/\sqrt{3} \tag{64}$$

$$\Rightarrow \varepsilon(\vec{k}) = 2t\left(\cos(k_x) + \cos(k_y) + \cos(k_x - k_y)\right) + \mu \tag{65}$$

The first Brillouin zone is $\{k_x, k_y\} \in [-\pi, \pi]^2$. The unit cell volume is $\Omega = \sqrt{3}/2$. Consistency check: $|\det(\mathrm{J})| = \Omega^{-1}$, since the substitution simply transforms the unti cell to a square of unit volume. So the integrals that have to be evaluated are of the form

$$\langle N_i \rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} f(\varepsilon(\vec{k}), \mu), \tag{66}$$

$$\langle \chi_{ij} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\Big[i\left(k_y, (1/\sqrt{3})(2k_x - k_y)\right) \cdot (\vec{R}_j - \vec{R}_i)\Big] f(\varepsilon(\vec{k}), \mu) \tag{67}$$

Check that under a particle hole transformation chi changes sign, sign of t is doing just that and this brings the *pi*-flux phase.
The condition is $\langle N_i \rangle \stackrel{!}{=} 1 \; \forall i$. By setting setting $t = -1$, the chemical potential is numerically determined in units of t using mathematica. I choose $t = -1$ because I anticipate t to be negative for a positive χ. (This may not be true depending on κ and χ, but as long as it is self-consistent evrything is fine.) For the numerical integration I used a Fermi-function with a small temperature ($\beta = 200t$) instead of a step-function, since the ladder is not analytical and causes problems for the numerical algorithm. The result is $\mu = -0.834737\ t$. The bond-parameters are calulcated the same way, the result is $\langle \chi_{ij} \rangle = 0.164713$.

Now, it is possible to express μ only in units of J. Use the formula

$$t_{ij} = -J\langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa\, |\langle \chi_{ij} \rangle|^2). \tag{68}$$

Set $\kappa = 10$:

$$t_{ij} = -0.8095\ J \tag{69}$$

$$\mu = -0.67574\ J \tag{70}$$

$$\langle \chi_{ij} \rangle = 0.164713 \tag{71}$$

The energy density (that is also the free energy desnity at $T = 0$) is

$$\frac{\langle H\rangle}{N} = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \varepsilon(\vec{k}) f(\varepsilon(\vec{k})) - \mu + 6\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta}\rangle|^2 \rangle\right) |\langle \chi_{i,i+\delta}\rangle|^2 \ = -0.888304\, J \tag{72}$$

Observtaion; the chemical potential drops out of the Free energy.

One can also directly calculate results for the state with all fluxes π by performing a particle-hole-transformation:

$$t \to -t$$
$$\mu \to -\mu \tag{73}$$
$$\Rightarrow \varepsilon(\vec{k}) \to -\varepsilon(\vec{k}) \tag{74}$$
$$\Rightarrow f(\varepsilon(\vec{k})) \to 1 - f(\varepsilon(\vec{k}) \tag{75}$$

It can be shown that

$$\chi \to -\chi. \tag{76}$$

The result is:

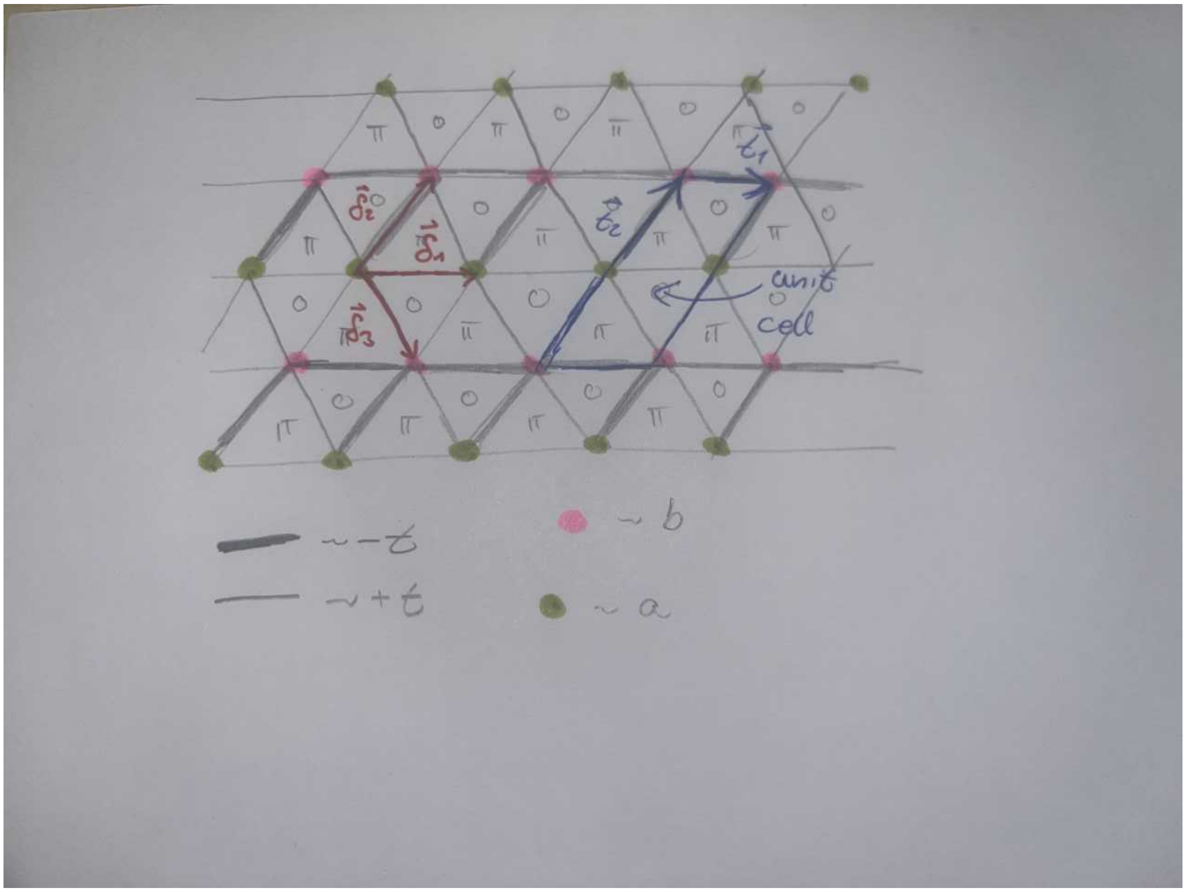
$$\frac{\langle H\rangle_\pi}{N} = -0.888302\, J \tag{77}$$

5.1.2 Pedro's method

The longer explenation for this method is in the next section. Here I will only say that the free energy is the same except for γ, which gets replaced by a factor ϕ (as mentioned above the chemical potentioal does not contribute to the free energy):

5.2 0-π and π-0 States

5.2.1 pedestrian method



The system is a triangular bravais lattice with a two-atomic basis. The primitve translations are

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \tag{78}$$

Hence, two types of lattice sites exist, a and b (see sketch). Also χ_{ij}, and thereby t_{ij} can change sign (see sketch). The Hamiltonian is

$$H = \frac{1}{2}\sum_{\langle i,j\rangle,\sigma}\left(ta^\dagger_{i\sigma}a_{j\sigma} - tb^\dagger_{i\sigma}b_{j\sigma} + t^{(ab)}_{ij}a^\dagger_{i\sigma}b_{j\sigma} + \text{h.c.}\right) + \mu_a\sum_{i,\sigma}\left(a^\dagger_{i\sigma}a_{i\sigma} - 1\right) + \mu_b\sum_{i,\sigma}\left(b^\dagger_{i\sigma}b_{i\sigma} - 1\right) + \text{const.} \tag{79}$$

*a and b sites are connectd by bonds of both flavours. Mind that the sum above generates 24 terms. The constant is the same as for the 0-flux case, provided that the absolute value of $\chi_{i,i+\delta}$ is uniform.

$$\frac{1}{2}\sum_{i,\delta}2\tilde{J}\left(1 - 6|\langle\chi_{i,i+\delta}|^2\rangle\right)|\langle\chi_{i,i+\delta}\rangle|^2 = 6N\tilde{J}\left(1 - 6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2 \tag{80}$$

This is not quite wright yet. The Fourier transformation has to be modified bc there are only N/2 of a and b-sites respectively

$$a_{i\sigma} = \sqrt{\frac{2}{N}}\sum_k e^{i\vec{k}\vec{R}_i}a_{k\sigma}, \tag{81}$$

and the same for $b_{i\sigma}$. The Hamiltonian (without the constants) becomes

$$H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t\cos\left(\vec{q}\vec{\delta}_1\right) + \mu_a & 2t\cos\left(\vec{q}\vec{\delta}_3\right) - 2it\sin\left(\vec{q}\vec{\delta}_2\right) \\ 2t\cos\left(\vec{q}\vec{\delta}_3\right) + 2it\sin\left(\vec{q}\vec{\delta}_2\right) & -2t\cos\left(\vec{q}\vec{\delta}_1\right) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \tag{82}$$

The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \tag{83}$$

.
The suitable transformation is

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1 & \sqrt{3} \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \tag{84}$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \tag{85}$$

$$\Rightarrow \det(J) = 1/\sqrt{3} \tag{86}$$

This yields

$$\cos\left(\vec{q}\vec{\delta}_1\right) \to \cos(k_y)$$

$$\cos\left(\vec{q}\vec{\delta}_2\right) \to \cos(k_y - k_x/2) \tag{87}$$

$$\sin\left(\vec{q}\vec{\delta}_3\right) \to \sin(k_x/2) \tag{88}$$

The dispersion relation is

$$\varepsilon(\vec{k})_\pm = \mu_a + \mu_b \pm \sqrt{(\mu_a - \mu_b)^2 + 2t\cos(k_y)(\mu_a - \mu_b) + 4t^2\cos^2(k_y) + 4t^2\cos^2(k_y - k_x/2) + 4t^2\sin^2(k_x/2)} \tag{89}$$

Apparently, half filling is achieved globally, iff $\mu_a = -\mu_b \equiv \mu$. (The spectrum is symmetric around $\varepsilon = 0$ in this case.) In order to fulfill the local constraint one still has to calculate the local expectation values. It would be weird however, if the chemical potential differed between a and b sites (-¿ charge density waves?).

$$\Rightarrow \varepsilon(\vec{k})_\pm = \pm 2|t|\sqrt{(\mu/(2t) + \cos(k_y))^2 + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \tag{90}$$

It is correct that ε only depends on the absolute value of t, it looks like this would completely neutrilise the particle-hole transformation.The periodicity is a bit weird here, since the expression is not periondic in 2π in every direction as one would expect. I am prettys rue this comes from the fact that the distance between NN is the same as for the 0-flux case but the Brilouin zone got cut in hoalf in one direction. represent hamiltonian as

$$H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \vec{h} \cdot \vec{\sigma} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix} \tag{91}$$

$$\vec{h} = \begin{pmatrix} 2t\cos(k_y - k_x/2)) \\ 2t\sin(k_x/2) \\ 2t\cos(k_y) + \mu \end{pmatrix}^T \tag{92}$$

Then $\varepsilon(\vec{q})_\pm = \pm|\vec{h}|$ and

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t\cos(k_y)+\mu)}}\begin{pmatrix} \varepsilon_+ + 2t\cos(k_y)+\mu & -2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) \\ 2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) & \varepsilon_+ + 2t\cos(k_y)+\mu \end{pmatrix} \tag{93}$$

is the transformation matrix U. It has the properties that it is unitary and has determinant one. The expectation values are more involved bc of said matrix transformation. For example

$$\begin{aligned}
\langle a^\dagger_{i\uparrow} b_{j\uparrow}\rangle &\overset{\mathrm{FT}}{=} \frac{2}{N}\sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{p}\vec{R}_i)} \underbrace{\langle a^\dagger_p b_q\rangle}_{\propto \delta_{q,p}} \\
&= \frac{2}{N}\sum_{q} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \langle a^\dagger_q b_q\rangle \\
&\overset{\mathrm{BT}}{=} \frac{2}{N}\sum_{q} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left(u^*_{11}u_{21}f(\varepsilon_+) + u^*_{12}u_{22}f(\varepsilon_-)\right)
\end{aligned} \tag{94}$$

Simplify this using

1. $\varepsilon_+ = -\varepsilon_- \Rightarrow f(\varepsilon_+) + f(\varepsilon_-) = 1$
2. $u_{11} \in \mathbb{R}$
3. $u_{11} = u_{22}$
4. $u^*_{12} = -u_{21}$
5. $|u_{11}|^2 + |u_{12}|^2 = 1$ (normalization of eigenvectors)

$$\begin{aligned}
u^*_{11}u_{21}f(\varepsilon_+) + u^*_{12}u_{22}f(\varepsilon_-) &\overset{1.}{=} (u^*_{11}u_{21} - u_{22}u^*_{12})f(\varepsilon_+) + u_{22}u^*_{12} \\
&\overset{2.,3.,4.}{=} 2u_{11}u_{21}f(\varepsilon_+) - u_{11}u_{21} \\
&= u_{11}u_{21}(2f(\varepsilon_+) - 1)
\end{aligned} \tag{95}$$

Due to the unit cell doubling the thermodynamic limit has to be slightly modified $\sum_k \to ((\Omega N)/2)/(2\pi)^2 \int dk$. (Each unit cell has 2 sites, this is why the factor 1/2 appears.)

$$\begin{aligned}
\Rightarrow \langle a^\dagger_{i\uparrow} b_{j\uparrow}\rangle &= \Omega \int\limits_{1.\mathrm{BZ}} \frac{d^2q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} u_{11}u_{21}(2f(\varepsilon_+) - 1) \\
&= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i\left(k_y, 1/\sqrt{3}(k_x - k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] u_{11}u_{21}(2f(\varepsilon_+) - 1)
\end{aligned} \tag{96}$$

$$u_{11}u_{21} = (t/\varepsilon)(\cos(k_y - k_x/2) + i\sin(k_x/2)) \tag{97}$$

Now for the other expectation values:

$$\Rightarrow \langle b^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i\left(k_y, 1/\sqrt{3}(k_x - k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] u_{11}u^*_{21}\,(2f(\varepsilon_+) - 1) \tag{98}$$

$$
\begin{aligned}
\langle a^\dagger_{i\uparrow} a_{j\uparrow}\rangle = ... &= \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[|u_{11}|^2 f(\varepsilon_+) + |u_{12}|^2 f(\varepsilon_-)\right] \\
&\overset{1.}{=} \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[|u_{12}|^2 - |u_{11}|^2) f(\varepsilon_-) + |u_{11}|^2\right] \\
&\overset{5.}{=} \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[(1-2u_{11}^2) f(\varepsilon_-) + u_{11}^2\right] \\
&\overset{1.}{=} \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[(1-2u_{11}^2)(1- f(\varepsilon_+)) + u_{11}^2\right]
\end{aligned} \tag{99}
$$

$$
\Rightarrow \langle a^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i\left(k_y, 1/\sqrt{3}(k_x-k_y)\right)^T(\vec{R}_j-\vec{R}_i)\right]\left[(1-2u_{11}^2)(1-f(\varepsilon_+)) + u_{11}^2\right] \tag{100}
$$

$$
\Rightarrow \langle b^\dagger_{i\uparrow} b_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i\left(k_y, 1/\sqrt{3}(k_x-k_y)\right)^T(\vec{R}_j-\vec{R}_i)\right]\left[(1-2u_{11}^2)f(\varepsilon_+) + u_{11}^2\right] \tag{101}
$$

$$
u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t\cos(k_y) + \mu) \tag{102}
$$

$$
\langle N^a\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}(1-2u_{11}^2)f(\varepsilon_-) + u_{11}^2 \tag{103}
$$

$$
\langle N^b\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}(1-2u_{11}^2)f(\varepsilon_+) + u_{11}^2 \tag{104}
$$

The constraint $\langle N_i^a\rangle = 1 = \langle N_i^b\rangle$ should, as gloal halffiling is already ensured and $\langle N^a\rangle$ and $\langle N^b\rangle$ are uniform by construction, be equivalent to $\langle N^a\rangle = \langle N^b\rangle$. Using the expressions above:

$$
\langle N^a\rangle - \langle N^b\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}(1-2u_{11}^2)(1-2f(\varepsilon_+)) \overset{!}{=} 0 \tag{105}
$$

Numerics say that $\langle N^a\rangle = \langle N^b\rangle$ for $\mu = 0$ (and therefore $\mu_a = \mu_b = 0$). I feel like there is a more suphisticated argument to be made here, mabye even without the use of mathematica. Maybe one can find some symmetry argument using the form of $1 - 2f(\varepsilon_+)$.

Sanity check:

$$
\langle N^a + N^b\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} 1 = 2 \tag{106}
$$

Calculating the mean field parameters is less straight forward, as the formulae also depend on the types of sites they connect. The formulae are

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} a_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1-2u_{11}^2)(1-f(\varepsilon_+)) + u_{11}^2\right] \tag{107}$$

$$\frac{1}{2}\sum_\sigma \langle b^\dagger_{i\sigma} b_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1-2u_{11}^2)f(\varepsilon_+) + u_{11}^2\right] \tag{108}$$

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_3\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y-k_x/2)} u_{11}u_{21}(2f(\varepsilon_+)-1) \tag{109}$$

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_2\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2} u_{11}u_{21}^*(2f(\varepsilon_+)-1) \tag{110}$$

Consistency check: wehn reversing the direction of the bonds the $\vec{\delta}_2$ bond paramter should obtain a minus sign but the $\vec{\delta}_3$ bond parameter not. I wasnt able to figure this out by just looking at the formulae, so Ill just have to check by looking at the numerics for now.

Another observation:

$$\begin{aligned}
\frac{1}{2}\sum_\sigma \left(\langle a^\dagger_{i\sigma} a_{i+\delta_1\sigma}\rangle + \langle b^\dagger_{i\sigma} b_{i+\delta_1\sigma}\rangle\right) &= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} \\
&= \frac{1}{2\pi}\int\limits_{[-\pi,\pi]} dk_y e^{ik_y} \\
&= \frac{\sin(\pi)}{\pi} \\
&= 0
\end{aligned} \tag{111}$$

$$\Rightarrow \frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} a_{i+\delta_1\sigma}\rangle = -\frac{1}{2}\sum_\sigma \langle b^\dagger_{i\sigma} b_{i+\delta_1\sigma}\rangle, \tag{112}$$

as expected. The numerical results are

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} a_{i+\delta_1\sigma}\rangle = 0.19754 \tag{113}$$

$$\frac{1}{2}\sum_\sigma \langle b^\dagger_{i\sigma} b_{i+\delta_1\sigma}\rangle = -0.19754 \tag{114}$$

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_3\sigma}\rangle = 0.19754 \tag{115}$$

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_2\sigma}\rangle = 0.19754 \tag{116}$$

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i-\delta_2\sigma}\rangle = -0.19754 \tag{117}$$

The corresponding hopping amplitudes are

$$t(\pm 0.19754) = \mp 0.87690J \tag{118}$$

$$\tag{119}$$

At last I will derive the energy density (since I had trouble with this before).

$$\begin{aligned}\frac{\langle H\rangle}{N} &= \frac{1}{N}\left\langle \sum_{q\sigma} \varepsilon_{+\sigma}\alpha^\dagger_{+\sigma}\alpha_{+\sigma} + \varepsilon_{-\sigma}\alpha^\dagger_{-\sigma}\alpha_{-\sigma}\right\rangle \\ &\overset{\text{th.lim.}}{\longrightarrow} 2\frac{1}{N}\frac{\Omega N}{2}\int\limits_{1.\mathrm{BZ}} \frac{d^2q}{(2\pi)^2}\varepsilon_+ f(\varepsilon_+) + \varepsilon_- f(\varepsilon_-) \\ &= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(2f(\varepsilon_+)-1)\end{aligned} \tag{120}$$

(the blue 2 in the second line is the spin degeneracy)
Observation: the energy density is invariant under particle-hole transformation

$$\varepsilon_+(2f(\varepsilon_+)-1) \overset{\text{ph-trafo}}{\longrightarrow} \varepsilon_-(2f(\varepsilon_-)-1) = -\varepsilon_+(1-2f(\varepsilon_+)) = \varepsilon_+(2f(\varepsilon_+)-1) \tag{121}$$

The full energy density is:

$$\frac{\langle H\rangle}{N} = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(\vec{k})(2f(\varepsilon_+(\vec{k}))-1) + 6\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2\right) \ = -1.2206\ J \tag{122}$$

5.2.2 Pedro's method

Here I apply Pedros method using the terms from my Hamiltonian. The free energy is

$$f_0 \equiv \frac{F_0(\kappa,\chi,T=0)}{N} = \frac{E_0}{N} + \frac{1}{N}\sum_{\vec{k}}\varepsilon(\vec{k}) \tag{123}$$

$$E_0 = 6N\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2\right) \tag{124}$$

$$\varepsilon(\vec{k}) = -2\left|-\tilde{J}\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\right|\sqrt{\cos^2(k_1)+\cos^2(k_3)+\sin^2(k_2)} \tag{125}$$

From now on I use the shorthand $|\langle\chi_{ij}\rangle| \equiv \chi$.

$$\begin{aligned}\Rightarrow f_0 &= \tilde{J}\left(6(1-6\tilde{\kappa}\chi^2)\chi^2 - 2\gamma\left|\chi(1-4\tilde{\kappa}\chi^2)\right|\right) \\ &= \tilde{J}\left(\mp 2\gamma\chi + 6\chi^2 \pm 8\gamma\tilde{\kappa}\chi^3 - 36\tilde{\kappa}\chi^4\right)\end{aligned} \tag{126}$$

with

$$\gamma = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\sqrt{\cos^2(k_y)+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \approx 1.18524 \tag{127}$$

compared to Pedros results there is also a factor 2, besides other problems, because Pedro calculates f/M $(f/2)$!

$$\frac{\partial f_0}{\partial\chi} = \tilde{J}\left(\pm 2\gamma - 12\chi\right)\left(12\tilde{\kappa}\chi^2 - 1\right) \qquad \frac{\partial^2 f_0}{\partial\chi^2} = \tilde{J}\left(12 \pm 48\gamma\tilde{\kappa}\chi - 432\tilde{\kappa}\chi^2\right) \tag{128}$$

The extremal points of the free energy are at

$$\chi' = \gamma/6, \qquad \chi'' = \pm 1/\sqrt{12\tilde{\kappa}} \tag{129}$$

$-1/\sqrt{12\tilde{\kappa}}$ is always a Maximum. $\gamma/6$ is a minimum for $\kappa \leq 6/\gamma^2$, afterwards $1/\sqrt{12\tilde{\kappa}}$ becomes a minimum:

$$f_0 = \begin{cases} \frac{\tilde{J}\gamma^2}{6}\left(\frac{\tilde{\kappa}\gamma^2}{18} - 1\right), & \tilde{\kappa} \leq 6/\gamma^2 \\ \tilde{J}\left(\frac{1}{4\tilde{\kappa}} - \frac{2\gamma}{9}\sqrt{\frac{3}{\tilde{\kappa}}}\right), & \tilde{\kappa} > 6/\gamma^2 \end{cases} \tag{130}$$

$6/\gamma^2 \approx 2.1355$. For $\kappa = 10 \Rightarrow \tilde{\kappa} = 1.\bar{6}$ and therefore $\tilde{\kappa} < 6/\gamma^2$. Hence, the free energy minimum is at

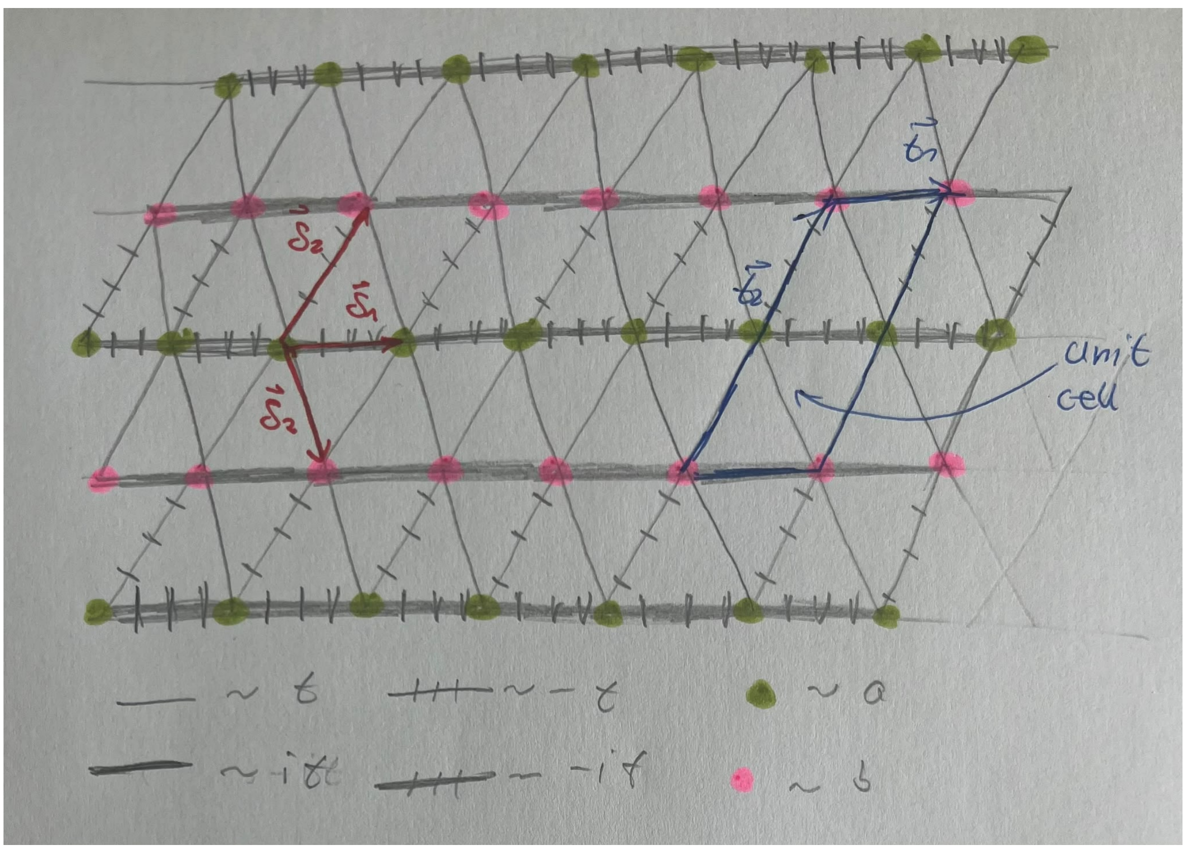
$$\chi = \gamma/6 = 0.19754, \tag{131}$$

with the free energy

$$f_0 = -1.22206\, J. \tag{132}$$

These correspond to the values I calculated using the pedestrian method above. Mind that under particle-hole transformation the calculation doesn't change (the reason basically being the absolute value in the energy, or more physically the particle-hole symmetry of the system).

5.3 $\pi/2$-$\pi/2$ and $-\pi/2$ - $-\pi/2$ States



5.3.1 Pedestrian Method

This is very similar to the $\pi - 0-$flux-state. The Bloch Hamiltonian in this case reads

$$H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t\sin\left(\vec{q}\vec{\delta}_1\right) + \mu_a & 2t\cos\left(\vec{q}\vec{\delta}_3\right) - 2it\sin\left(\vec{q}\vec{\delta}_2\right) \\ 2t\cos\left(\vec{q}\vec{\delta}_3\right) + 2it\sin\left(\vec{q}\vec{\delta}_2\right) & -2t\sin\left(\vec{q}\vec{\delta}_1\right) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \tag{133}$$

So the only difference is $\cos\left(\vec{q}\vec{\delta}_1\right) \to \sin\left(\vec{q}\vec{\delta}_1\right)$. Therefore, I will skip ahead in the calculations:
Maybe I should show explicitly that $\mu = 0$, I guess $\mu_a = -\mu_b = \mu$ is clear

$$\mu_a = \mu_b = 0, \tag{134}$$
$$\varepsilon(\vec{k})_\pm = \pm 2|t|\sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \tag{135}$$

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t\sin(k_y))}} \begin{pmatrix} \varepsilon_+ + 2t\sin(k_y) & -2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) \\ 2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) & \varepsilon_+ + 2t\sin(k_y) \end{pmatrix} \tag{136}$$

$$u_{11}u_{21} = (t/\varepsilon_+)(\cos(k_y - k_x/2) + i\sin(k_x/2)) \tag{137}$$
$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t\sin(k_y)) \tag{138}$$

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} a_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2\right] \tag{139}$$
$$\frac{1}{2}\sum_\sigma \langle b^\dagger_{i\sigma} b_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1 - 2u_{11}^2)f(\varepsilon_+) + u_{11}^2\right] \tag{140}$$
$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_3\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11}u_{21}(2f(\varepsilon_+) - 1) \tag{141}$$
$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_2\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2} u_{11}u_{21}^*(2f(\varepsilon_+) - 1) \tag{142}$$

One has to be a bit careful of where the sign of t enters when choosing $t = 1$ for the self consistency. Due to the complex conjugation the σ_z-terms are insensitive to the sign of t, but the (real) σ_x- and σ_y-terms are not. The results (choosing $t = 1$ are):

$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} a_{i+\delta_1\sigma}\rangle = -0.200169i \tag{143}$$
$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} a_{i-\delta_1\sigma}\rangle = 0.200169i \tag{144}$$
$$\frac{1}{2}\sum_\sigma \langle b^\dagger_{i\sigma} b_{i+\delta_1\sigma}\rangle = 0.200169i \tag{145}$$
$$\frac{1}{2}\sum_\sigma \langle b^\dagger_{i\sigma} b_{i-\delta_1\sigma}\rangle = -0.200169i \tag{146}$$
$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_3\sigma}\rangle = 0.200169 \tag{147}$$
$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i-\delta_3\sigma}\rangle = 0.200169 \tag{148}$$
$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i+\delta_2\sigma}\rangle = 0.200169 \tag{149}$$
$$\frac{1}{2}\sum_\sigma \langle a^\dagger_{i\sigma} b_{i-\delta_2\sigma}\rangle = -0.200169 \tag{150}$$

This complies with the cartoon on top. The value for t is

$$|t_{ij}| = \left| J\langle\chi_{ij}\rangle^*(1+\kappa/2 - 4\kappa|\langle\chi_{ij}\rangle|^2 \right| \overset{\kappa=10}{=} 0.88020\,J. \tag{151}$$

The free energy is

$$\frac{\langle H\rangle}{N} = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(\vec{k})(2f(\varepsilon_+(\vec{k}))-1) + 6\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2\right) \ = -1.24978\ J \tag{152}$$

5.3.2 Pedro's Method

The expression for the free energy is identical to the $0-\pi$-state except that γ gets replaced by a different constant α:

$$\alpha = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\sqrt{\sin^2(k_y)+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \approx 1.20102 \tag{153}$$

Crucally α is larger than gamma. The results for this method are

$$|\chi| = \alpha/6 = 0.20017, \tag{154}$$

$$f_0 = \tilde{J}\left(\frac{1}{4\tilde{\kappa}} - \frac{2\alpha}{9}\sqrt{\frac{3}{\tilde{\kappa}}}\right) = -1.24979\,J \tag{155}$$

The methods agree.

Part III
other

6. mail 10.07

$$\mathcal{H} = \frac{J_H}{M'} \sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j + \frac{\kappa}{(M')^2} \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \tag{156}$$

transformations:

$$\text{Eric}: \; \vec{S}_i = \sum_{\alpha\beta} f^\dagger_{i\alpha} \frac{\vec{\sigma}}{2} f_{i\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{1}{4} - \frac{1}{2} \sum_{\alpha\beta} f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{i\beta} \tag{157}$$

$$\text{Pedro}: \; S_i^{\beta\alpha} = f^\dagger_{i\beta} f_{i\alpha} - \frac{Q}{M} \delta_{\alpha\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{M}{4} - \sum_{\alpha\beta} f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{i\beta} \tag{158}$$

difference of factor 2 for M = 2, which is my case. (Also, for me $M' = 1$ while Pedro used $M' = M$.) mean field Hamiltonains:

$$\text{Eric}: \mathcal{H} \approx -J \sum_{\langle ij \rangle} \Big[(1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \left(f^\dagger_{j\uparrow} f_{i\uparrow} + f^\dagger_{j\downarrow} f_{i\downarrow} \right) + \text{h.c.} \tag{159}$$

$$+ 2(-1 - \kappa/2 + 6\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \Big] \tag{160}$$

$$\text{Pedro}: \mathcal{H} \approx -J \sum_{\langle ij \rangle} \Big[2 + \boxed{2\kappa} - 16\kappa |\langle \chi_{ij} \rangle|^2) \langle \chi_{ij} \rangle \left(f^\dagger_{j\uparrow} (f_{i\uparrow} + f^\dagger_{j\downarrow} f_{i\downarrow} \right) + \text{h.c.} \tag{161}$$

$$+ 2(-2 \boxed{-2\kappa} + 24\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \Big] \tag{162}$$

Every term without a κ (coming from the bilinear exchange term) is off by a factor of 2. Every term with a κ (coming from the biquadratic exchange term) is off by a factor of 4. Hence, the only problem is the factor 2 in the fermionic representation. The boxed terms dont appear in Pedros notes as they go like M^{-1} and are therfore irrelevant in the large M limit.